Scoring Point-Cloud Distributional Submissions

The deconvolution incentive, jittered outcomes, and the heat ladder

Peter Cotton · *Working draft v0.2* · 2026

Status. An evolving working note, not a finished paper. Theorems 1–3 are proved below at working-draft rigor in the population-law setting: the cloud is idealised by its sampling law, and the smoothing/jitter kernel is fixed in advance. Finite clouds and participant-endogenous bandwidths are outside the proved results (§7). The mechanism and the closed forms are implemented and unit-tested in `mechanisms/nearest_the_pin.py` (`mollified_log_score`, and tests that verify Theorem 1 numerically in both directions). The prior-art audit lives in `research/mollified-scoring-and-the-heat-ladder.md`. *v0.2:* sharpened hypotheses following referee feedback — conditional truthfulness gap (Thm. 1), integrability assumptions (Thm. 2), exogenous bandwidth requirement, standard de Bruijn regularity in place of a tail overclaim (Thm. 3), and an explicit level-vs-difference payment analysis with a new proposition that band scores are strictly proper. Drafted with AI assistance; errors are mine.

Abstract

Forecasting contests often accept a cloud of Monte-Carlo samples, smooth it into a kernel density estimate, and score the result at the realised outcome. This design is improper: the optimal cloud is drawn from the deconvolution of the forecaster's belief by the smoothing kernel, which for a Gaussian belief means shaving the squared bandwidth off the variance (Theorem 1). Jittering the outcome with the same kernel restores strict propriety, because convolution by a kernel whose characteristic function has dense support is injective on probability measures (Theorem 2). Repeating the repaired score across smoothing scales decomposes the log-score edge, via de Bruijn's identity, into Fisher-divergence payments for shape plus a coarse-scale payment for mass (Theorem 3), yielding a budget-balanced multiscale pool. All results are population-level, with the kernel fixed in advance.

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1. The problem

A contest accepts from each participant i a cloud $x_1^{(i)}, \dots, x_m^{(i)} \in \mathbb{R}^d$, smooths it into a density $\hat{q}_i = \frac{1}{m} \sum_j \varphi_h(\cdot - x_j^{(i)})$, and — when the outcome z is revealed — rewards $\log \hat{q}_i(z)$ (or splits a pot in proportion to $\hat{q}_i(z)$, the exponentiated form). This is the reward rule of the nearest-the-pin parimutuel and of monteprediction-style contests, and the standard evaluation protocol for sample-based generative models.

The question this paper answers: *what cloud should a rational participant submit?* The answer is not "samples from your belief," and the failure has a closed form, a one-line repair, and, once repaired, a multi-scale structure with its own uses.

Throughout, we idealise the cloud by its sampling law ρ ($m \to \infty$; finite m is Open Problem 1), so the mechanism observes $T_h\rho := \rho * \varphi_h$, where φ_h is a density on $\mathbb{R}^d$, symmetric, strictly positive (the Gaussian $N(0, h^2 I)$ is canonical). The truth is p^*; all expectations below are assumed finite (for Gaussian kernels and beliefs with finite second moments, $\log T_h\rho$ has at-worst-quadratic tails, so this is mild).

2. The deconvolution incentive

Theorem 1 (improperness of raw-outcome KDE scoring). *Let $S_{\text{raw}}(\rho, z) = \log(T_h\rho)(z)$. Then*

$$\mathbb{E}_{z\sim p^*} S_{\text{raw}}(\rho, z) \;=\; -H(p^*) \;-\; \mathrm{KL}(p^* \,\|\, T_h\rho),$$

so the report solves $\min_{q\in T_h\mathcal{P}} \mathrm{KL}(p^*\|q)$, where $\mathcal{P}$ is the set of probability laws. Consequently:

(i) If the deconvolution $\rho^\dagger := T_h^{-1}p^*$ exists in $\mathcal{P}$, it is the unique optimal report (uniqueness by the injectivity of T_h, Lemma 1), it differs from the truth, and truthful reporting is strictly suboptimal with truthfulness gap

$$\Delta \;=\; \mathrm{KL}(p^*\|p^* * \varphi_h) \;>\; 0.$$

(ii) If $p^* \notin T_h\mathcal{P}$, the optimum is the KL projection of p^* onto the convex class $T_h\mathcal{P}$, and Δ above is only the loss of the truthful report relative to the unattainable ideal $q = p^*$; the gap to the attainable optimum is $\Delta - \inf_{\rho\in\mathcal{P}} \mathrm{KL}(p^*\|T_h\rho)$, which can vanish in degenerate cases (for a point-mass belief the truthful cloud is optimal).

(iii) Gaussian closed form: $p^* = N(\mu, \Sigma)$, $\varphi_h = N(0, h^2 I)$. The deconvolution exists iff $\Sigma - h^2 I \succeq 0$, and then $\rho^\dagger = N(\mu, \Sigma - h^2 I)$. In one dimension with $\tau^2 > h^2$: shave h^2 off the variance, with gap $\Delta = \frac{1}{2}[\log(1 + h^2/\tau^2) - h^2/(\tau^2 + h^2)] \approx h^4/(4\tau^4)$.

Proof. The identity is the standard cross-entropy decomposition, $\int p^* \log q \;=\; -H(p^*) - \mathrm{KL}(p^*\|q)$, applied to $q = T_h\rho$; maximizing over ρ is minimizing KL over the convex image class $T_h\mathcal{P}$. (i) If $p^* \in T_h\mathcal{P}$ the KL term can be driven to zero, its unique minimum, and only by $q = p^*$; Lemma 1 lifts uniqueness of q to uniqueness of ρ; the truthful report attains $\mathrm{KL}(p^*\|p_h^*)$ where $p_h^* := p^* * \varphi_h$, and $p_h^* \neq p^*$ always: taking characteristic functions, $\widehat{p}^*(\omega)\,(1 - \widehat{\varphi}_h(\omega)) = 0$ for all ω; for the Gaussian kernel $\widehat{\varphi}_h(\omega) = e^{-h^2|\omega|^2/2} < 1$ off $\omega = 0$, so $\widehat{p}^*$ would have to vanish off the origin, impossible for a characteristic function (continuity and $\widehat{p}^*(0) = 1$). (ii) KL is jointly lower semicontinuous and strictly convex in its second argument where finite, and $T_h\mathcal{P}$ is convex, so the attainable optimum is the KL projection when attained; the point-mass example ($p^* = \delta_\mu$: the truthful cloud maximises $(T_h\rho)(\mu)$) shows the gap to the attainable optimum can be zero. (iii) Gaussians: convolution adds covariances; the KL between $N(0, \tau^2)$ and $N(0, \tau^2 + h^2)$ is the stated expression; Taylor expansion gives $h^4/(4\tau^4)$. ■

Remarks. (a) When the deconvolution fails to exist ($\Sigma - h^2 I \not\succeq 0$, or a rough p^*), the optimum is the KL projection of p^* onto $T_h\mathcal{P}$ and can degenerate toward atomic clouds — exactly the point-mass exploit of Theis, van den Oord & Bethge (2016), who observed the improperness (without the deconvolution characterization). (b) The gap is fourth order in h, which is why the flaw survives casual inspection — but it is a *slope*, and any optimising submitter, human or fitted, walks down it. (c) The phenomenon is the log-score/bandwidth sibling of the "fair scores" findings for the ensemble CRPS (Fricker, Ferro & Stephenson 2013; Ferro 2014), where the analogous cheat is derived along the ensemble-size axis.

3. The repair: jitter the pin

Definition (mollified log score). With $\varepsilon \sim N(0, I)$,

$$S_h(\rho, z) \;=\; \mathbb{E}_\varepsilon\big[\log(T_h\rho)(z + h\varepsilon)\big] \;=\; (\varphi_h * \log T_h\rho)(z).$$

Operationally: *settle at a jittered outcome* $z' = z + h\varepsilon$ — or use the right-hand form, which integrates the jitter analytically and makes settlement deterministic. The two have the same expectation, hence identical incentive properties.

Lemma 1 (injectivity criterion). *Convolution T_φ is injective on probability measures iff the zero set of the characteristic function has empty interior,*

$$\operatorname{int}\{\omega : \hat{\varphi}(\omega) = 0\} \;=\; \emptyset$$

(equivalently, $\{\hat{\varphi} \neq 0\}$ is dense in $\mathbb{R}^d$).

Proof. ($\Leftarrow$) $T_\varphi\rho = T_\varphi\rho'$ gives $(\hat{\rho} - \hat{\rho}')\hat{\varphi} \equiv 0$, so $\hat{\rho} = \hat{\rho}'$ on a dense set, hence everywhere by continuity of characteristic functions, hence $\rho = \rho'$ by the uniqueness theorem. ($\Rightarrow$) If $\hat{\varphi}$ vanishes on an open ball B (with $0 \notin B$, since $\hat{\varphi}(0) = 1$), take $\psi \in C_c^\infty(B)$, $\psi \neq 0$, and set $\hat{g}(\omega) = \psi(\omega) + \overline{\psi(-\omega)}$, so that g is a real Schwartz function (a finite signed density) with total mass $\int g = \hat{g}(0) = 0$ and $\hat{g}$ supported where $\hat{\varphi} = 0$. Choose a base density p with tails heavier than Schwartz decay (Cauchy), so $|\epsilon g| \leq p$ pointwise for small $\epsilon > 0$. Then $q = p + \epsilon g$ is a probability density distinct from p, and $\widehat{T_\varphi q} = \hat{\varphi}\,(\hat{p} + \epsilon\hat{g}) = \hat{\varphi}\,\hat{p}$: two distinct laws with identical smoothings. ■

This is Wiener-flavoured (Wiener's Tauberian theorem is the L^1 statement); for probability measures the continuity of characteristic functions buys the dense-support version. Practical reading: Gaussian and Laplace jitter are injective (nonvanishing characteristic functions); the uniform kernel is also fine (sinc zeros are isolated, hence the nonvanishing set is dense); band-limited kernels fail (e.g. Fejér-type kernels, whose characteristic function vanishes on a half-line — two beliefs agreeing on low frequencies become indistinguishable).

Theorem 2 (kernel-channel properness). *Let S be a scoring rule, K a Markov kernel with push-forward T_K on laws, and define*

$$S_K(\rho, z) \;=\; \mathbb{E}_{z' \sim K(z,\cdot)}\big[S(T_K\rho,\, z')\big].$$

Assume (a) $\mathbb{E}_{w \sim T_K p}\, |S(T_K\rho, w)| < \infty$ for the laws p, ρ under comparison (or adopt an extended-expectation convention), and (b) $T_K\mathcal{P}$ lies within the report class on which S is defined. If S is proper, S_K is proper. If S is strictly proper on $T_K\mathcal{P}$, then S_K is strictly proper on $\mathcal{P}$ iff T_K is injective on $\mathcal{P}$. In particular the mollified log score S_h is strictly proper for the pre-smoothing law whenever the zero set of $\hat{\varphi}$ has empty interior — for Gaussian jitter, always.

Proof. By Fubini, $\mathbb{E}_{z\sim p^*}S_K(\rho,z) = \mathbb{E}_{w\sim T_K p^*}S(T_K\rho, w)$: the expected score of the report $T_K\rho$ for the outcome law $T_K p^*$, under S. Properness of S maximizes this at $T_K\rho = T_K p^*$, which the truthful $\rho = p^*$ achieves — properness. Strict properness of S on the image class forces $T_K\rho = T_K p^*$ at any maximizer; injectivity of T_K lifts this to $\rho = p^*$, and conversely if T_K is not injective two distinct reports tie. For $S = \log$ and K = convolution by φ_h: the log score is strictly proper on densities, $T_K\rho = \rho * \varphi_h$ is a density, and Lemma 1 gives injectivity. ■

How much to jitter? Exactly as much as you smooth. The jitter is not a free parameter; it is pinned, kernel for kernel, to the mechanism's own smoothing. Jittering with s.d. j while smoothing with bandwidth h pairs the jittered truth $p^* * \varphi_j$ with the smoothed report $\rho * \varphi_h$, and in the all-Gaussian setting the optimal report variance is

$$v^* \;=\; \tau^2 + j^2 - h^2.$$

At $j = 0$ this is Theorem 1's shave. At $j = h$, and only at $j = h$, the optimum is the truth. At $j > h$ the mechanism pays padding by $j^2 - h^2$. The formula holds where admissible ($v^* \geq 0$); otherwise the Gaussian-family optimum sits on the boundary (the KL-projection case of Theorem 1(ii)). Under-jitter rewards sharpening, over-jitter rewards blurring, and matching the bandwidth is the unique fixed point (the general statement is Theorem 2 with the *same* kernel on both sides; a mismatched pair elicits $\arg\min_\rho \mathrm{KL}(p^* * \varphi_j \,\|\, \rho * \varphi_h)$, the j-blurred belief deconvolved by h).

The bandwidth must be exogenous. Theorem 2 assumes a *fixed* channel: the smoothing/jitter kernel must not depend on the submission being scored. A data-driven bandwidth computed *from the participant's own cloud* (Scott's rule on the submission, as reference KDE implementations default to) puts $h(\rho)$ under the participant's control, the score becomes $\mathbb{E}_\varepsilon \log(T_{h(\rho)}\rho)(z + h(\rho)\varepsilon)$, and strict properness no longer follows from Theorem 2. In a contest, freeze the bandwidth before submissions are observed (from the outcome history, a reference climatology, or a posted rule) and jitter with that frozen kernel. The incentives of participant-endogenous bandwidths are an open problem (§7).

Attribution. The properness of the convolved score against a noised outcome is not new: Bröcker & Smith (2007, §5) prove it for a general observation-noise channel, and Ferro (2017, Prop. 3) works out exactly the white-noise/Gaussian case. What both left open is strictness. Bröcker & Smith, verbatim: "If S is strictly proper though, $\bar{S}$ is not necessarily strictly proper, because if $\bar{q}(z) = \bar{p}(z)$, this does not necessarily mean equality of $p(x)$ and $q(x)$." Lemma 1 closes exactly that gap. The discrete-outcome analog is the label-noise literature's forward loss correction with an invertible transition matrix (Patrini et al. 2017, Thm. 2; van Rooyen & Williamson 2018). Verification treats outcome noise as a nuisance to be endured, and Ferro explicitly doubts the "efficacy" of perturbing observations; read as mechanism design, jitter matched to the mechanism's own smoothing is what makes the point-cloud game strictly proper.

4. The heat ladder

Let $p_t := p * N(0, tI)$, the heat flow ($\partial_t p_t = \frac{1}{2}\Delta p_t$). To keep one time variable, write $u = h^2 + t$ for the *total* smoothing scale: the rung at flow time t is the §3 score with kernel $\varphi_{\sqrt{u}}$ — for clouds this is free to compute, since it is *the same cloud* scored with bandwidth $\sqrt{u}$ against a pin jittered by the same kernel. Below, subscripts t refer to the flow started from the already h-smoothed laws, so every rung the mechanism runs has $u \geq h^2 > 0$.

Theorem 3 (scale decomposition of the log-score edge). *Let p^*, ρ have finite second moments, and suppose the standard relative de Bruijn regularity conditions hold on $[s, T]$: $\mathrm{KL}(p_t^* \| \rho_t)$ finite along the flow, the relative Fisher divergence integrable on the interval, and enough decay to justify the integrations by parts below. Then for $0 \leq s < T$,*

$$\frac{d}{dt}\,\mathrm{KL}(p_t^* \| \rho_t) \;=\; -\tfrac{1}{2}\, D_F(p_t^* \| \rho_t), \qquad \mathrm{KL}(p_s^* \| \rho_s) = \mathrm{KL}(p_T^* \| \rho_T) + \tfrac{1}{2}\int_s^T D_F(p_t^* \| \rho_t)\, dt,$$

where $D_F(p\|q) = \int p\, \|\nabla \log p - \nabla \log q\|^2$ is the Fisher divergence.

Proof. Write $p = p_t^*$, $q = \rho_t$, both solving $\partial_t u = \frac{1}{2}\Delta u$. Then

$$\frac{d}{dt}\int p \log\frac{p}{q} \;=\; \int (\partial_t p)\log\frac{p}{q} \;+\; \underbrace{\int \partial_t p}_{=0} \;-\; \int \frac{p}{q}\,\partial_t q.$$

First term: $\frac{1}{2}\int \Delta p\; \log(p/q) = -\frac{1}{2}\int \nabla p \cdot \nabla \log(p/q) = -\frac{1}{2}\int p\, \nabla \log p \cdot (\nabla \log p - \nabla \log q)$. Third term: $-\frac{1}{2}\int \frac{p}{q}\,\Delta q = \frac{1}{2}\int \nabla\!\left(\frac{p}{q}\right)\cdot \nabla q = \frac{1}{2}\int p\,(\nabla \log p - \nabla \log q)\cdot \nabla \log q$. Summing,

$$\frac{d}{dt}\,\mathrm{KL} = -\tfrac{1}{2}\int p\, \|\nabla \log p - \nabla \log q\|^2 = -\tfrac{1}{2} D_F(p\|q),$$

and integrating over $[s, T]$ gives the display. After smoothing by $u \geq h^2 > 0$ both densities are positive and smooth, which is necessary but not by itself sufficient: Gaussian convolution does *not* replace heavy tails by Gaussian tails, so the vanishing of boundary terms is an assumption (the stated regularity), automatic for the Gaussian and compactly supported cases used in the examples. ■

The differential identity is de Bruijn's, in relative form (Stam 1959; Barron 1986; Lyu 2009); its integral form prices the likelihood of diffusion models (Song, Durkan, Murray & Ermon 2021). Note the bandwidth floor does real work: at $t = 0$ an empirical cloud has no density and the identity is vacuous; every rung the mechanism actually runs starts at $t \geq h^2$, where everything is smooth.

The heat-ladder pool. Fix scales $0 = t_0 < t_1 < \cdots < t_K = T$ and weights $w_k \geq 0$. Participant i stakes s_i and submits one cloud. Two payment schedules must be distinguished, because they buy different things.

Level schedule. Rung k splits $w_k \sum_i s_i$ by a budget-balanced rule driven by the rung score $S_i^{(k)} := S_{\sqrt{h^2+t_k}}(\rho_i, z)$ — the stake-weighted additive form $\Delta W_i^{(k)} = w_k\, s_i\,(S_i^{(k)} - \bar{S}^{(k)})$ with $\bar{S}^{(k)}$ the stake-weighted mean (Lambert et al. 2008). Each rung is strictly proper (Theorem 2), so any nonnegative-weighted sum is. But the expected-regret decomposition carries *cumulative* weights on the Fisher bands, not the rung weights themselves: writing $\mathrm{KL}_k := \mathrm{KL}(p_{t_k}^* \| \rho_{t_k})$,

$$\sum_{k=0}^{K} w_k \,\mathrm{KL}_k \;=\; \Big(\sum_{k=0}^{K} w_k\Big)\mathrm{KL}_K \;+\; \tfrac{1}{2}\sum_{\ell=0}^{K-1}\Big(\sum_{k=0}^{\ell} w_k\Big)\int_{t_\ell}^{t_{\ell+1}} D_F(p_t^*\|\rho_t)\,dt.$$

Difference schedule. To pay a Fisher band directly, use the score *differences* as the payment driver.

Proposition (band scores are strictly proper). *The difference score $S^{(k)} - S^{(k+1)}$ has expected regret $\mathrm{KL}_k - \mathrm{KL}_{k+1} = \frac{1}{2}\int_{t_k}^{t_{k+1}} D_F(p_t^*\|\rho_t)\,dt \geq 0$, with equality iff $\rho = p^*$. It is therefore itself a strictly proper score, even though a difference of proper scores is not proper in general.*

Proof. The expected value of $S^{(k)}$ under p^* is $-H(p_{t_k}^*) - \mathrm{KL}_k$; the entropy terms are report-independent, so the regret of the difference is $\mathrm{KL}_k - \mathrm{KL}_{k+1}$, which is the band integral by Theorem 3. It vanishes iff $D_F(p_t^*\|\rho_t) = 0$ on the band, i.e. $\nabla \log \rho_t = \nabla \log p_t^*$ p_t^*-a.e.; after Gaussian smoothing both densities are everywhere positive, so the log-ratio is constant, and both integrating to one forces $\rho_t = p_t^*$, hence $\rho = p^*$ by Lemma 1. ■

Corollary. *(i) Each rung (or band), hence the tower, is budget-balanced in the additive stake-weighted form. (ii) Each level score is strictly proper for the cloud law (Theorem 2) and each band score is strictly proper (the Proposition), so truthful submission is optimal in the small-stake, risk-neutral limit; the multiplicative pot split requires a log-wealth (Kelly) model stated separately. (iii) Band payments purchase (integrated) Fisher divergence, which is Hyvärinen-scored shape, invariant to the normalization of the submission; the top level pays $\mathrm{KL}(p_T^*\|\rho_T)$, which at mode-connecting scales carries the between-mode* mass *that score matching is blind to (Wenliang & Kanagawa 2020; Zhang et al. 2022; Koehler, Heckett & Risteski 2023). Under the level schedule the band weights are the cumulative sums above.*

Practicalities: rung scores are positively correlated (one cloud, re-smoothed), so discriminative value concentrates in a few well-separated scales; K of order 3–5, geometrically spaced, mirrors the noise ladders of annealed score matching (Song & Ermon 2019).

5. Historical note: the jitter was intuition first

The microprediction platform (Cotton 2022; launched 2019) added a small amount of noise to submissions and to the ground truth before settling its cloud-based lotteries. This was merely intuitive, a fairness-and-anti-gaming instinct about discreteness and ties, with no incentive theorem attached; the platform paper recorded the practice in one line and moved on. The verification literature, meanwhile, treated outcome noise as a defect: something to be modelled away (Saetra et al. 2004; Candille & Talagrand 2008), with Ferro (2017) explicitly doubting the value of perturbing observations.

Theorem 2 is the missing theorem: symmetrized jitter is what makes a smoothed-submission game strictly proper for the submitted law. The theory also sharpens the intuition into design guidance it could not supply on its own: the jitter distribution matters. Kernels whose characteristic function has dense support (Gaussian, Laplace, even uniform) preserve strict properness; band-limited kernels do not (Lemma 1). Intuition chose jitter; the theorem chooses *which* jitter.

6. Related work

Improperness of sample-based scoring: Theis, van den Oord & Bethge (2016) (KDE log-likelihood "an improper scoring function"); the fair-scores line for the ensemble CRPS (Bröcker

2012; Fricker, Ferro & Stephenson 2013; Ferro 2014); the estimator view of KDE log scores (Krüger, Lerch, Thorarinsdottir & Gneiting 2021); discrete impossibility and randomized repair (Kimpara, Frongillo & Waggoner 2023). Convolved scores under observation error: Bröcker & Smith (2007); Ferro (2017); Bessac & Naveau (2021). Noisy-channel learning: Patrini et al. (2017); van Rooyen & Williamson (2018); surrogate scoring rules (Liu, Wang & Chen 2022). Transform-properness: Allen, Ginsbourger & Ziegel (2023, Prop. 4 published / Prop. 3 arXiv v1); Pic, Dombry, Naveau & Taillardat (2025, Prop. 1) — deterministic injective transforms; Theorem 2 is the Markov-kernel extension. The identity: Stam (1959); Barron (1986); Lyu (2009); Song et al. (2021). Local scores: Hyvärinen (2005); Parry, Dawid & Lauritzen (2012). Nearest mechanism neighbours, each missing a leg: Lang, Leutbecher & Maciel (2025) — a Gaussian scale-ladder of proper scores as a *training loss*; Dudík, Wang, Pennock & Rothschild (2021) — a multi-resolution *market* by partition refinement, subsidized rather than budget-balanced; Lambert et al. (2008, 2015) and the microprediction pool — self-funding cloud wagering at a single scale. A full audit with verdicts is in the companion research note.

7. Open problems

1. **Fair rungs.** The finite-m correction making each rung's expected score optimized by *sampling* from the belief (the log-score/KDE analog of Ferro's fair CRPS), and its interaction with the jitter. All results here are population-level; the finite-cloud game is not covered by the theorems.
2. **Endogenous bandwidth.** If the bandwidth is computed from the participant's own submission (Scott's rule on the cloud), the channel is no longer fixed and Theorem 2 does not apply; the participant controls both ρ and $h(\rho)$. Characterize the optimal joint deviation, and whether any self-referential bandwidth rule preserves truthfulness.
3. **Optimal scale weights.** For wealth-concentration objectives, is there a closed-form optimal $w(t)$ — and does it recover the likelihood weighting of Song et al. (2021)?
4. **Anisotropic rungs.** Reference-covariance flows in place of isotropic heat, preserving rung-wise strict properness (cf. the anisotropic sliced scores note).
5. **The Kelly interpolation.** Between $b \to 0$ (linear pot split, mode-seeking) and $b = 1$ (log score, truthful), characterize the rung-wise optimal misreport as a function of the stake fraction.

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